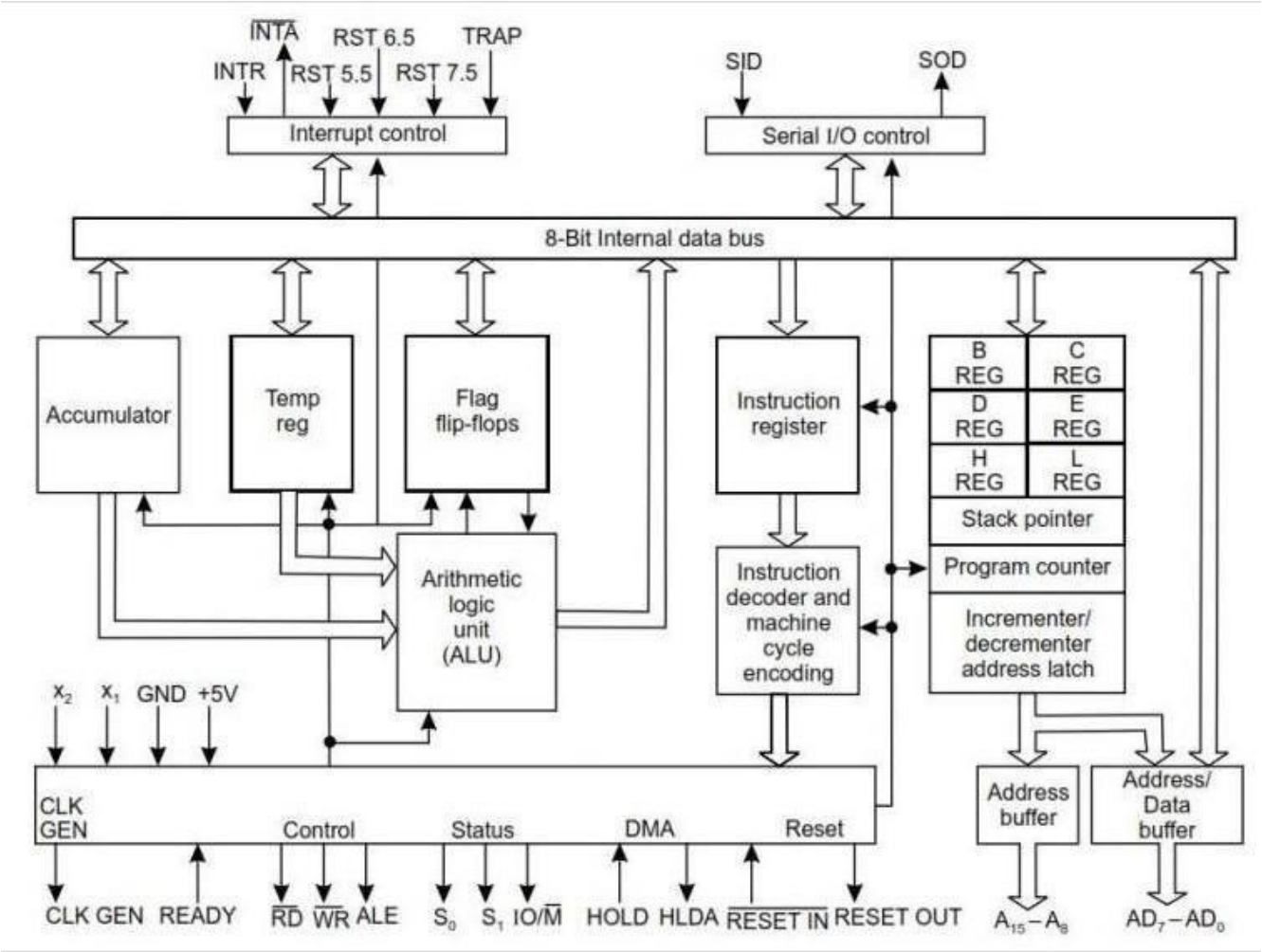
Architecture of Microprocessor 8085

Intel 8085 is an 8- bit Microprocessor. It is a 40 pin Integrated circuit on a single LSI chip. Its architecture is shown below. 8085 consists of three main sections, an ALU, Timing & control unit & set of registers.



1. Arithmetic & Logic Unit

The ALU performs different arithmetic and logical operations such as addition, subtraction, logical AND, OR & Ex-OR etc. It takes an operand from Accumulator & stores the result into it.

1.1 Flags

Microprocessor 8085 has five flip-flops, which reflects the results of operations performed by ALU. They are called Zero (Z), Carry (CY), Sign (S), Parity (P), and Auxiliary Carry (AC) flags. The most commonly used flags are Zero, Carry, and Sign.

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
S	Z	-	AC	-	P	-	CY

Flag register of 8085

1. When an arithmetic operation produces a carry, the carry flag (CY) sets to 1, else resets to 0. It is set & reset in case of addition as well as subtraction.
2. When the result of an arithmetic or logical operation contains even number of 1's, the parity flag (P) sets to 1, else resets to 0.
3. When there is a carry out of the bit number 3 to the bit number 4, in case of an arithmetic operation, the auxiliary carry flag (AC) sets to 1, else resets to 0.
4. When an arithmetic or logical operation produces result as 0, the zero flag (Z) sets to 1, else resets to 0.
5. When the result of an arithmetic or logical operation is negative, the sign flag (S) sets to 1, else resets to 0.

2. Registers

The 8085 has six general purpose registers & an accumulator. In addition, it has two 16-bit registers: the stack pointer and the program counter. The six general-purpose registers are used to store 8-bit data & are identified as B, C, D, E, H, and L. They can be combined as register pairs - BC, DE, and HL - to perform some 16-bit operations. The H-L pair is also used as memory pointer.

2.1 Accumulator (A)

The accumulator is an 8-bit register that is a part of arithmetic & logic unit (ALU). This register is used to store 8-bit data. It serves as one input to the ALU, & the result is stored in the accumulator.

2.2 Program Counter (PC)

The program counter, a 16-bit register is used to sequence the execution of the instructions. It points to the memory location from which the next byte is to be fetched. When a byte (machine code) is being fetched, the program counter is incremented by one to point to the next memory location.

2.3 Stack Pointer (SP)

The stack pointer, a 16-bit register is used as a memory pointer. It points to a memory location in R/W memory, called the stack. The stack pointer holds the address of top element of data stored in stack.

3. Instruction Register & Decoder

The instruction register stores the current instruction of a program temporarily. Decoder then takes these instructions and 'decodes' or interprets them. Decoded instructions are then passed to the next stage.

4. Control Unit

The control unit supervises the entire operation of the microprocessor, by generating signals that are necessary for execution of instructions. It controls the data flow between CPU & peripherals. It also provides status, control & timing signals which are required for the operation of memory & I/O devices.

PROGRAM 1(a): An Assembly Language Program (ALP) to perform Addition of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	67				MOV	H,A
8004	3A	00	81		LDA	8100
8007	84				ADD	H
8008	32	02	81		STA	8102
800B	76				HLT	

[8100]=52 **52 + 38 = 8A** **[8102]=8A**

[8101]=38

PROGRAM 1(b): An Assembly Language Program (ALP) to perform Subtraction of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	67				MOV	H,A
8004	3A	00	81		LDA	8100
8007	94				SUB	H
8008	32	02	81		STA	8102
800B	76				HLT	

[8100]=52 ` **52 - 38 = 1A** **[8102]=1A**

[8101]=38

PROGRAM 2: An Assembly Language Program (ALP) to perform Decimal Addition of two 2-digit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	67				MOV	H,A
8004	3A	00	81		LDA	8100
8007	84				ADD	H
8008	27				DAA	
8009	32	02	81		STA	8102
800C	76				HLT	

[8100]=52

[8101]=38

$(52)_{10} + (38)_{10} = (90)_{10}$

[8102]=90



PROGRAM 3(a): An Assembly Language Program (ALP) to perform Addition of two 16-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	82		LDA	8200
8003	67				MOV	H,A
8004	3A	00	81		LDA	8100
8007	84				ADD	H
8008	32	00	83		STA	8300
800B	3A	01	82		LDA	8201
800E	5F				MOV	E,A
800F	3A	01	81		LDA	8101
8012	8B				ADC	E
8013	32	01	83		STA	8301
8016	76				HLT	

[8101]=38 [8100]=72 3872 + 2593 = 5E05 [8301]=5E [8300]=05
[8201]=25 [8200]=93

PROGRAM 3(b): An Assembly Language Program (ALP) to perform Subtraction of two 16-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	82		LDA	8200
8003	67				MOV	H,A
8004	3A	00	81		LDA	8100
8007	94				SUB	H
8008	32	00	83		STA	8300
800B	3A	01	82		LDA	8201
800E	5F				MOV	E,A
800F	3A	01	81		LDA	8101
8012	9B				SBB	E
8013	32	01	83		STA	8301
8016	76				HLT	

[8101]=38 [8100]=72 3872 - 2593 = 12DF [8301]=12 [8300]=DF
[8201]=25 [8200]=93

PROGRAM 4: An Assembly Language Program (ALP) to perform Addition of two 16-bit numbers using DAD instruction.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	2A	00	81		LHLD	8100
8003	EB				XCHG	
8004	2A	00	82		LHLD	8200
8007	19				DAD	D
8008	22	00	83		SHLD	8300
800B	76				HLT	

[8101]=38 [8100]=72 3872 + 2593 = 5E05 [8301]=5E [8300]=05
[8201]=25 [8200]=93



PROGRAM 5(a): An Assembly Language Program (ALP) to perform Addition of elements of two arrays.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	11	00	81		LXI	D,8100
8003	21	00	82		LXI	H,8200
8006	0E	05			MVI	C,05
8008	1A			BACK:	LDAX	D
8009	86				ADD	M
800A	77				MOV	M,A
800B	23				INX	H
800C	13				INX	D
800D	0D				DCR	C
800E	C2	08	80		JNZ	BACK
8011	76				HLT	

[8100]=70	**[8200]=25**	**70 + 25 = 95**	**[8200]=95**
[8101]=68	**[8201]=26**	**68 + 26 = 8E**	**[8201]=8E**
[8102]=50	**[8202]=29**	**50 + 29 = 79**	**[8202]=79**
[8103]=56	**[8203]=28**	**56 + 28 = 7E**	**[8203]=7E**
[8104]=51	**[8204]=27**	**51 + 27 = 78**	**[8204]=78**

PROGRAM 5(b): An Assembly Language Program (ALP) to perform Subtraction of elements of an array from another.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	11	00	81		LXI	D,8100
8003	21	00	82		LXI	H,8200
8006	0E	05			MVI	C,05
8008	1A			BACK:	LDAX	D
8009	96				SUB	M
800A	77				MOV	M,A
800B	23				INX	H
800C	13				INX	D
800D	0D				DCR	C
800E	C2	08	80		JNZ	BACK
8011	76				HLT	

[8100]=70 [8200]=25 70 - 25 = 4B [8200]=4B

[8101]=68 [8201]=26 68 - 26 = 42 [8201]=42

[8102]=50 [8202]=29 50 - 29 = 27 [8202]=27

[8103]=56 [8203]=28 56 - 28 = 2E [8203]=2E

[8104]=51 [8204]=27 51 - 27 = 2A [8204]=2A

PROGRAM 6: An Assembly Language Program (ALP) to perform Addition of two arrays of decimal numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	11	00	81		LXI	D,8100
8003	21	00	82		LXI	H,8200
8006	0E	05			MVI	C,05
8008	1A			BACK:	LDAX	D
8009	86				ADD	M
800A	27				DAA	
800B	77				MOV	M,A
800C	23				INX	H
800D	13				INX	D
800E	0D				DCR	C
800F	C2	08	80		JNZ	BACK
8012	76				HLT	

[8100]=70 **[8200]=25** $(70)_{10} + (25)_{10} = (95)_{10}$ **[8200]=95**

[8101]=68 **[8201]=26** $(68)_{10} + (26)_{10} = (94)_{10}$ **[8201]=94**

[8102]=50 **[8202]=29** $(50)_{10} + (29)_{10} = (79)_{10}$ **[8202]=79**

[8103]=56 **[8203]=28** $(56)_{10} + (28)_{10} = (84)_{10}$ **[8203]=84**

[8104]=51 **[8204]=27** $(51)_{10} + (27)_{10} = (78)_{10}$ **[8204]=78**

PROGRAM 7: An Assembly Language Program (ALP) to transfer data from an array to another.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	11	00	81		LXI	D,8100
8003	21	00	82		LXI	H,8200
8006	0E	05			MVI	C,05
8008	1A			BACK:	LDAX	D
8009	77				MOV	M,A
800A	23				INX	H
800B	13				INX	D
800C	0D				DCR	C
800D	C2	08	80		JNZ	BACK
8010	76				HLT	

[8100]=01

[8101]=02

[8102]=03

[8103]=04

[8104]=05

[8200]=01

[8201]=02

[8202]=03

[8203]=04

[8204]=05

PROGRAM 8: An Assembly Language Program (ALP) to add the binary numbers in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	21	00	81		LXI	H,8100
8003	06	05			MVI	B,05
8005	3E	00			MVI	A,00
8007	0E	00			MVI	C,00
8009	86			BACK:	ADD	M
800A	D2	0E	80		JNC	SKIP
800D	0C				INR	C
800E	23			SKIP:	INX	H
800F	05				DCR	B
8010	C2	09	80		JNZ	BACK
8013	32	00	82		STA	8200
8016	79				MOV	A,C
8017	32	01	82		STA	8201
801A	76				HLT	

[8100]=55 **55 + 66 + 49 + 58 + 67 = 1C3** **[8201]=01 [8200]=C3**

[8101]=66

[8102]=49

[8103]=58

[8104]=67

PROGRAM 9: An Assembly Language Program (ALP) to add the decimal numbers in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	21	00	81		LXI	H,8100
8003	06	05			MVI	B,05
8005	3E	00			MVI	A,00
8007	0E	00			MVI	C,00
8009	86			BACK:	ADD	M
800A	27				DAA	
800B	D2	0F	80		JNC	SKIP
800E	0C				INR	C
800F	23			SKIP:	INX	H
8010	05				DCR	B
8011	C2	09	80		JNZ	BACK
8014	32	00	82		STA	8200
8017	79				MOV	A,C
8018	32	01	82		STA	8201
801B	76				HLT	

[8100]=55

[8101]=66

[8102]=49

[8103]=58

[8104]=67

$(55)_{10} + (66)_{10} + (49)_{10} + (58)_{10} + (67)_{10} = (295)_{10}$

[8201]=02 [8200]=95

PROGRAM 10(a): An Assembly Language Program (ALP) to count the Negative values in an array of signed numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	05			MVI	C,05
8002	21	00	81		LXI	H,8100
8005	06	00			MVI	B,00
8007	7E			BACK:	MOV	A,M
8008	17				RAL	
8009	D2	0D	80		JNC	SKIP
800C	04				INR	B
800D	23			SKIP:	INX	H
800E	0D				DCR	C
800F	C2	07	80		JNZ	BACK
8012	78				MOV	A,B
8013	32	00	82		STA	8200
8016	76				HLT	

[8100]=26

[8101]=82

[8102]=49

[8103]=28

[8104]=90

82→ 10000010 →Negative

90→ 10010000 → Negative

[8200]=02

PROGRAM 10(b): An Assembly Language Program (ALP) to count the Positive values in an array of signed numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	05			MVI	C,05
8002	21	00	81		LXI	H,8100
8005	06	00			MVI	B,00
8007	7E			BACK:	MOV	A,M
8008	17				RAL	
8009	DA	0D	80		JC	SKIP
800C	04				INR	B
800D	23			SKIP:	INX	H
800E	0D				DCR	C
800F	C2	07	80		JNZ	BACK
8012	78				MOV	A,B
8013	32	00	82		STA	8200
8016	76				HLT	

[8100]=26

[8101]=82

[8102]=49

[8103]=28

[8104]=90

26→ 00100110 →Positive

49→ 01001001 → Positive

28→ 00101000 → Positive

[8200]=03

PROGRAM 11(a): An Assembly Language Program (ALP) to count the Odd numbers in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	05			MVI	C,05
8002	21	00	81		LXI	H,8100
8005	06	00			MVI	B,00
8007	7E			BACK:	MOV	A,M
8008	1F				RAR	
8009	D2	0D	80		JNC	SKIP
800C	04				INR	B
800D	23			SKIP:	INX	H
800E	0D				DCR	C
800F	C2	07	80		JNZ	BACK
8012	78				MOV	A,B
8013	32	00	82		STA	8200
8016	76				HLT	

[8100]=26 **49→ 01001001 →Odd** **[8200]=01**

[8101]=82

[8102]=49

[8103]=28

[8104]=90

PROGRAM 11(b): An Assembly Language Program (ALP) to count the Even numbers in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	05			MVI	C,05
8002	21	00	81		LXI	H,8100
8005	06	00			MVI	B,00
8007	7E			BACK:	MOV	A,M
8008	1F				RAR	
8009	DA	0D	80		JC	SKIP
800C	04				INR	B
800D	23			SKIP:	INX	H
800E	0D				DCR	C
800F	C2	07	80		JNZ	BACK
8012	78				MOV	A,B
8013	32	00	82		STA	8200
8016	76				HLT	

[8100]=26

[8101]=82

[8102]=49

[8103]=28

[8104]=90

26→ 00100110 → Even

82→ 10000010 → Even

28→ 00101000 → Even

90→ 10010000 → Even

[8200]=04

PROGRAM 12: An Assembly Language Program (ALP) to perform Subtraction of two 8-bit numbers using 2's Complement.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	2F				CMA	
8004	3C				INR	A
8005	67				MOV	H,A
8006	3A	00	81		LDA	8100
8009	84				ADD	H
800A	32	02	81		STA	8102
800D	76				HLT	

[8100]=52 **52 - 38 = 1A** **[8102]=1A**

[8101]=38

PROGRAM 13: An Assembly Language Program (ALP) to perform Decimal Subtraction of two 2-digit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	47				MOV	B,A
8004	3E	99			MVI	A,99
8006	90				SUB	B
8007	3C				INR	A
8008	4F				MOV	C,A
8009	3A	00	81		LDA	8100
800C	81				ADD	C
800D	27				DAA	
800E	32	02	81		STA	8102
8011	76				HLT	

[8100]=52
[8101]=38

$(52)_{10} - (38)_{10} = (14)_{10}$

[8102]=14

PROGRAM 14: An Assembly Language Program (ALP) to swap the two hexadecimal digits in an 8-bit number.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	81		LDA	8100
8003	0F				RRC	
8004	0F				RRC	
8005	0F				RRC	
8006	0F				RRC	
8007	32	00	82		STA	8200
800A	76				HLT	

[8100]=52 **52→25** **[8200]=25**



PROGRAM 15(a): An Assembly Language Program (ALP) to count number of 1's in an 8-bit number.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	08			MVI	C,08
8002	3A	00	81		LDA	8100
8005	06	00			MVI	B,00
8007	07			BACK:	RLC	
8008	D2	0C	80		JNC	SKIP
800B	04				INR	B
800C	0D			SKIP:	DCR	C
800D	C2	07	80		JNZ	BACK
8010	78				MOV	A,B
8011	32	01	81		STA	8101
8014	76				HLT	

[8100]= 57 **57→ 01010111** **[8101]=05**

PROGRAM 15(b): An Assembly Language Program (ALP) to count number of 0's in an 8-bit number.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	08			MVI	C,08
8002	3A	00	81		LDA	8100
8005	06	00			MVI	B,00
8007	07			BACK:	RLC	
8008	DA	0C	80		JC	SKIP
800B	04				INR	B
800C	0D			SKIP:	DCR	C
800D	C2	07	80		JNZ	BACK
8010	78				MOV	A,B
8011	32	01	81		STA	8101
8014	76				HLT	

[8100]= 57 **57→ 01010111** **` [8101]=03**

PROGRAM 16: An Assembly Language Program (ALP) to reverse the bit positions in an 8-bit number.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	0E	08			MVI	C,08
8002	3A	00	81		LDA	8100
8005	47				MOV	B,A
8006	16	00			MVI	D,00
8008	78			BACK:	MOV	A,B
8009	17				RAL	
800A	47				MOV	B,A
800B	7A				MOV	A,D
800C	1F				RAR	
800D	57				MOV	D,A
800E	0D				DCR	C
800F	C2	08	80		JNZ	BACK
8012	7A				MOV	A,D
8013	32	00	82		STA	8200
8016	76				HLT	

[8100]=54 54=01010100 → 00101010=2A [8200]=2A

PROGRAM 17: An Assembly Language Program (ALP) to perform Multiplication of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	4F				MOV	C,A
8004	3A	00	81		LDA	8100
8007	47				MOV	B,A
8008	AF				XRA	A
8009	57				MOV	D,A
800A	80			BACK:	ADD	B
800B	D2	0F	80		JNC	SKIP
800E	14				INR	D
800F	0D			SKIP:	DCR	C
8010	C2	0A	80		JNZ	BACK
8013	32	00	82		STA	8200
8016	7A				MOV	A,D
8017	32	01	82		STA	8201
801A	76				HLT	

[8100]=56 **56 X 21 = B16** **[8201]=0B [8200]=16**

[8101]=21

PROGRAM 18: An Assembly Language Program (ALP) to perform Decimal Multiplication of two 2-digit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	4F				MOV	C,A
8004	3A	00	81		LDA	8100
8007	47				MOV	B,A
8008	AF				XRA	A
8009	57				MOV	D,A
800A	80			BACK:	ADD	B
800B	27				DAA	
800C	D2	10	80		JNC	SKIP
800F	14				INR	D
8010	0D			SKIP:	DCR	C
8011	C2	0A	80		JNZ	BACK
8014	32	00	82		STA	8200
8017	7A				MOV	A,D
8018	32	01	82		STA	8201
801B	76				HLT	

[8100]=85 **$(85)_{10}$ X $(09)_{10}$ = $(765)_{10}$** **[8201]=07 [8200]=65**

[8101]=09

PROGRAM 19: An Assembly Language Program (ALP) to perform Division of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	01	81		LDA	8101
8003	47				MOV	B,A
8004	3A	00	81		LDA	8100
8007	0E	00			MVI	C,00
8009	B8			BACK:	CMP	B
800A	DA	12	80		JC	EXIT
800D	90				SUB	B
800E	0C				INR	C
800F	C3	09	80		JMP	BACK
8012	32	00	82	EXIT:	STA	8200
8015	79				MOV	A,C
8016	32	01	82		STA	8201
8019	76				HLT	

[8100]=31

$$\frac{31}{03} = 10 + \frac{01}{03}$$

[8200]=01 → (Remainder)

[8101]=03

[8201]=10 → (Quotient)

PROGRAM 20(a): An Assembly Language Program (ALP) to find Larger of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	81		LDA	8100
8003	47				MOV	B,A
8004	3A	01	81		LDA	8101
8007	B8				CMP	B
8008	D2	0C	80		JNC	SKIP
800B	78				MOV	A,B
800C	32	00	82	SKIP:	STA	8200
800F	76				HLT	

[8100]=56 **82 > 56** **[8200]=82**

[8101]=82

PROGRAM 20(b): An Assembly Language Program (ALP) to find Smaller of two 8-bit numbers.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	81		LDA	8100
8003	47				MOV	B,A
8004	3A	01	81		LDA	8101
8007	B8				CMP	B
8008	DA	0C	80		JC	SKIP
800B	78				MOV	A,B
800C	32	00	82	SKIP:	STA	8200
800F	76				HLT	

[8100]=56 **56 < 82** **[8200]=56**

[8101]=82

PROGRAM 21(a): An Assembly Language Program (ALP) to find the Largest number in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	21	00	81		LXI	H,8100
8003	0E	04			MVI	C,04
8005	7E				MOV	A,M
8006	23			BACK:	INX	H
8007	BE				CMP	M
8008	D2	0C	80		JNC	SKIP
800B	7E				MOV	A,M
800C	0D			SKIP:	DCR	C
800D	C2	06	80		JNZ	BACK
8010	32	00	82		STA	8200
8013	76				HLT	

[8100]=14

[8101]=10

[8102]=35

[8103]=29

[8104]=19

Maximum (14, 10, 35, 29, 19) = 35

[8200]=35

PROGRAM 21(b): An Assembly Language Program (ALP) to find the Smallest number in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	21	00	81		LXI	H,8100
8003	0E	04			MVI	C,04
8005	7E				MOV	A,M
8006	23			BACK:	INX	H
8007	BE				CMP	M
8008	DA	0C	80		JC	SKIP
800B	7E				MOV	A,M
800C	0D			SKIP:	DCR	C
800D	C2	06	80		JNZ	BACK
8010	32	00	82		STA	8200
8013	76				HLT	

[8100]=14

[8101]=10

[8102]=35

[8103]=29

[8104]=19

Minimum (14, 10, 35, 29, 19) = 10

[8200]=10

PROGRAM 22(a): An Assembly Language Program (ALP) to sort the elements of an array in Ascending order.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	16	04			MVI	D,04
8002	21	00	81	NEXT:	LXI	H,8100
8005	4A				MOV	C,D
8006	7E			BACK:	MOV	A,M
8007	23				INX	H
8008	BE				CMP	M
8009	DA	11	80		JC	SKIP
800C	46				MOV	B,M
800D	77				MOV	M,A
800E	2B				DCX	H
800F	70				MOV	M,B
8010	23				INX	H
8011	0D			SKIP:	DCR	C
8012	C2	06	80		JNZ	BACK
8015	15				DCR	D
8016	C2	02	80		JNZ	NEXT
8019	76				HLT	

[8100]=14
[8101]=10
[8102]=35
[8103]=29
[8104]=19

10 < 14 < 19 < 29 < 35

[8100]=10
[8101]=14
[8102]=19
[8103]=29
[8104]=35

PROGRAM 22(b): An Assembly Language Program (ALP) to sort the elements of an array in Descending order.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	16	04			MVI	D,04
8002	21	00	81	NEXT:	LXI	H,8100
8005	4A				MOV	C,D
8006	7E			BACK:	MOV	A,M
8007	23				INX	H
8008	BE				CMP	M
8009	D2	11	80		JNC	SKIP
800C	46				MOV	B,M
800D	77				MOV	M,A
800E	2B				DCX	H
800F	70				MOV	M,B
8010	23				INX	H
8011	0D			SKIP:	DCR	C
8012	C2	06	80		JNZ	BACK
8015	15				DCR	D
8016	C2	02	80		JNZ	NEXT
8019	76				HLT	

[8100]=14
[8101]=10
[8102]=35
[8103]=29
[8104]=19

35 > 29 > 19 > 14 > 10

[8100]=35
[8101]=29
[8102]=19
[8103]=14
[8104]=10

PROGRAM 23: An Assembly Language Program (ALP) to search for a byte in an array.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3A	00	82		LDA	8200
8003	21	00	81		LXI	H,8100
8006	0E	05			MVI	C,05
8008	06	FF			MVI	B,FF
800A	BE			BACK:	CMP	M
800B	CA	15	80		JZ	EXIT
800E	23				INX	H
800F	0D				DCR	C
8010	C2	0A	80		JNZ	BACK
8013	06	00			MVI	B,00
8015	78			EXIT:	MOV	A,B
8016	32	00	83		STA	8300
8019	76				HLT	

[8100]=14

[8101]=10

[8102]=35

[8103]=29

[8104]=19

i) [8200]= 45 (45 is not found in the array) → [8300]= 00
ii) [8200]= 35 (35 is found in the array) → [8300]= FF

PROGRAM 24(a): An Assembly Language Program (ALP) to generate an up-going Saw tooth waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	00			MVI	A,00
8006	D3	40		AGAIN:	OUT	P_A
8008	D3	41			OUT	P_B
800A	3C				INR	A
800B	C3	06	80		JMP	AGAIN

OUTPUT:

PROGRAM 24(b): An Assembly Language Program (ALP) to generate a down-going Saw tooth waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	00			MVI	A,00
8006	D3	40		AGAIN:	OUT	P_A
8008	D3	41			OUT	P_B
800A	3D				DCR	A
800B	C3	06	80		JMP	AGAIN

OUTPUT:

PROGRAM 25: An Assembly Language Program (ALP) to generate a Triangular waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	00			MVI	A,00
8006	D3	40		UP:	OUT	P_A
8008	D3	41			OUT	P_B
800A	3C				INR	A
800B	FE	FF			CPI	FF
800D	C2	06	80		JNZ	UP
8010	D3	40		DOWN:	OUT	P_A
8012	D3	41			OUT	P_B
8014	3D				DCR	A
8015	FE	00			CPI	00
8017	C2	10	80		JNZ	DOWN
801A	C3	06	80		JMP	UP

OUTPUT:

PROGRAM 26: An Assembly Language Program (ALP) to generate a Square waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	00		AGAIN:	MVI	A,00
8006	D3	40			OUT	P_A
8008	D3	41			OUT	P_B
800A	CD	20	80		CALL	DELAY
800D	3E	FF			MVI	A,FF
800F	D3	40			OUT	P_A
8011	D3	41			OUT	P_B
8013	CD	20	80		CALL	DELAY
8016	C3	04	80		JMP	AGAIN

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	06	FF		DELAY:	MVI	B,FF
8022	05			BACK:	DCR	B
8023	C2	22	80		JNZ	BACK
8026	C9				RET	

OUTPUT:

PROGRAM 27(a): An Assembly Language Program (ALP) to generate an up-going Staircase waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	00		AGAIN:	MVI	A,00
8006	D3	40		UP:	OUT	P_A
8008	D3	41			OUT	P_B
800A	CD	20	80		CALL	DELAY
800D	C6	33			ADI	33
800F	FE	FF			CPI	FF
8011	C2	06	80		JNZ	UP
8014	C3	04	80		JMP	AGAIN

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	06	FF		DELAY:	MVI	B,FF
8022	05			BACK:	DCR	B
8023	C2	22	80		JNZ	BACK
8026	C9				RET	

OUTPUT:

PROGRAM 27(b): An Assembly Language Program (ALP) to generate a down-going Staircase waveform by interfacing DAC with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	FF		AGAIN:	MVI	A,FF
8006	D3	40		DOWN:	OUT	P_A
8008	D3	41			OUT	P_B
800A	CD	20	80		CALL	DELAY
800D	DE	33			SUI	33
800F	FE	00			CPI	00
8011	C2	06	80		JNZ	DOWN
8014	C3	04	80		JMP	AGAIN

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	06	FF		DELAY:	MVI	B,FF
8022	05			BACK:	DCR	B
8023	C2	22	80		JNZ	BACK
8026	C9				RET	

OUTPUT:

PROGRAM 28(a): An Assembly Language Program (ALP) to rotate a Stepper motor in Clockwise direction by interfacing it with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	88			MVI	A,88
8006	D3	40		AGAIN:	OUT	P_A
8008	0F				RRC	
8009	F5				PUSH	PSW
800A	CD	20	80		CALL	DELAY
800D	F1				POP	PSW
800E	C3	06	80		JMP	AGAIN

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	11	FF	7F	DELAY:	LXI	D,7FFF
8023	1B			BACK:	DCX	D
8024	7A				MOV	A,D
8025	B3				ORA	E
8026	C2	23	80		JNZ	BACK
8029	C9				RET	

OUTPUT:

PROGRAM 28(b): An Assembly Language Program (ALP) to rotate a Stepper motor in Anti-Clockwise direction by interfacing it with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	3E	88			MVI	A,88
8006	D3	40		AGAIN:	OUT	P_A
8008	07				RLC	
8009	F5				PUSH	PSW
800A	CD	20	80		CALL	DELAY
800D	F1				POP	PSW
800E	C3	06	80		JMP	AGAIN

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	11	FF	7F	DELAY:	LXI	D,7FFF
8023	1B			BACK:	DCX	D
8024	7A				MOV	A,D
8025	B3				ORA	E
8026	C2	23	80		JNZ	BACK
8029	C9				RET	

OUTPUT:

PROGRAM 29: An Assembly Language Program (ALP) to control traffic lights at a Cross Road by interfacing traffic light module with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	0E	05		REPEAT:	MVI	C,05
8006	21	00	81		LXI	H,8100
8009	7E			NEXT:	MOV	A,M
800A	D3	40			OUT	P_A
800C	23				INX	H
800D	7E				MOV	A,M
800E	D3	41			OUT	P_B
8010	23				INX	H
8011	7E				MOV	A,M
8012	D3	42			OUT	P_C
8014	23				INX	H
8015	06	14			MVI	B,14
8017	CD	50	80		CALL	DELAY
801A	7E				MOV	A,M
801B	D3	40			OUT	P_A
801D	23				INX	H
801E	7E				MOV	A,M
801F	D3	41			OUT	P_B
8021	23				INX	H
8022	7E				MOV	A,M
8023	D3	42			OUT	P_C
8025	23				INX	H
8026	06	06			MVI	B,06
8028	CD	50	80		CALL	DELAY
802B	0D				DCR	C
802C	C2	09	80		JNZ	NEXT
802F	C3	04	80		JMP	REPEAT

DELAY SUBROUTINE:

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8050	11	FF	FF	DELAY:	LXI	D,FFFF
8053	1B			BACK:	DCX	D
8054	7A				MOV	A,D
8055	B3				ORA	E
8056	C2	53	80		JNZ	BACK
8059	05				DCR	B
805A	C2	50	80		JNZ	DELAY
805D	C9				RET	

INPUT:

[8100]=88 **[8101]=83** **[8102]=F2**

[8103]=88 **[8104]=84** **[8105]=F0**

[8106]=38 **[8107]=88** **[8108]=F4**

[8109]=48 **[810A]=88** **[810B]=F0**

[810C]=83 **[810D]=88** **[810E]=F8**

[810F]=84 **[8110]=88** **[8111]=F0**

[8112]=88 **[8113]=38** **[8114]=F1**

[8115]=88 **[8116]=48** **[8117]=F0**

[8118]=88 **[8119]=88** **[811A]=00**

[811B]=88 **[811C]=88** **[811D]=00**

OUTPUT:

PROGRAM 30: An Assembly Language Program (ALP) to display your Name and Roll number on LCD by interfacing LCD module with 8085.

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8000	3E	80			MVI	A,80
8002	D3	43			OUT	CWR
8004	CD	20	80		CALL	LCDINIT
8007	3E	80			MVI	A,80
8009	CD	50	80		CALL	LCDCMD
800C	21	00	81		LXI	H,8100
800F	CD	70	80		CALL	LCDSTR
8012	3E	C0			MVI	A,C0
8014	CD	50	80		CALL	LCDCMD
8017	21	00	82		LXI	H,8200
801A	CD	70	80		CALL	LCDSTR
801D	76				HLT	

SUBROUTINE 1: LCDINIT

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8020	3E	38		LCDINIT:	MVI	A,38
8022	CD	50	80		CALL	LCDCMD
8025	3E	0C			MVI	A,0C
8027	CD	50	80		CALL	LCDCMD
802A	3E	06			MVI	A,06
802C	CD	50	80		CALL	LCDCMD
802F	3E	01			MVI	A,01
8031	CD	50	80		CALL	LCDCMD
8034	C9				RET	

SUBROUTINE 2: DELAY

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8040	06	FF		DELAY:	MVI	B,FF
8042	05			BACK:	DCR	B
8043	C2	42	80		JNZ	BACK

8046	C9				RET	

SUBROUTINE 3: LCDCMD

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8050	D3	40		LCDCMD:	OUT	P_A
8052	3E	FC			MVI	A,FC
8054	D3	41			OUT	P_B
8056	CD	40	80		CALL	DELAY
8059	3E	F8			MVI	A,F8
805B	D3	41			OUT	P_B
805D	C9				RET	

SUBROUTINE 4: LCDCHR

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8060	D3	40		LCDCHR:	OUT	P_A
8062	3E	FD			MVI	A,FD
8064	D3	41			OUT	P_B
8066	CD	40	80		CALL	DELAY
8069	3E	F9			MVI	A,F9
806B	D3	41			OUT	P_B
806D	C9				RET	

SUBROUTINE 5: LCDSTR

MEMORY ADDRESS	MACHINE CODE			LABEL	INSTRUCTION MNEMONIC	OPERAND
8070	7E			LCDSTR:	MOV	A,M
8071	FE	00			CPI	00
8073	C8				RZ	
8074	CD	60	80		CALL	LCDCHR
8077	23				INX	H
8078	C3	70	80		JMP	LCDSTR

INPUT:

[8100]=	**[8200]=31**
[8101]=	**[8201]=36**
[8102]=	**[8202]=30**
[8103]=	**[8203]=34**
[8104]=	**[8204]=2D**
[8105]=	**[8205]=32**
[8106]=	**[8206]=33**
[8107]=	**[8207]=2D**
[8108]=	**[8208]=37**
[8109]=	**[8209]=33**
[810A]=	**[820A]=37**
[810B]=	**[820B]=2D**
[810C]=	**[820C]=**
[810D]=	**[820D]=**
[810E]=	**[820E]=**
[810F]=	**[820F]=00**

OUTPUT: